# Moss Distribution in the LEO1 Bath: Topographic Controls and Evidence for a Bounded Moisture Optimum

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**Abstract**

This technical note analyzes the spatial distribution of moss in the LEO1 bath using three rectified stitched images, a soil-surface digital elevation model, and a container-geometry raster. The analysis is centered on the biological pattern itself, rather than on the visually prominent drainage network alone. Across the three dates, the moss footprint is largely persistent, whereas the November-to-December change is interpreted mainly as seasonal greening of previously occupied regions. The strongest stable moss signal is displaced away from the deepest central trough and is concentrated on off-center shoulder zones. This pattern is consistent with a bounded moisture optimum: moss is favored in zones that remain moist, but not in zones that are maximally wet, strongly flushed, or hydrologically unstable.

## 1 Objective and Data

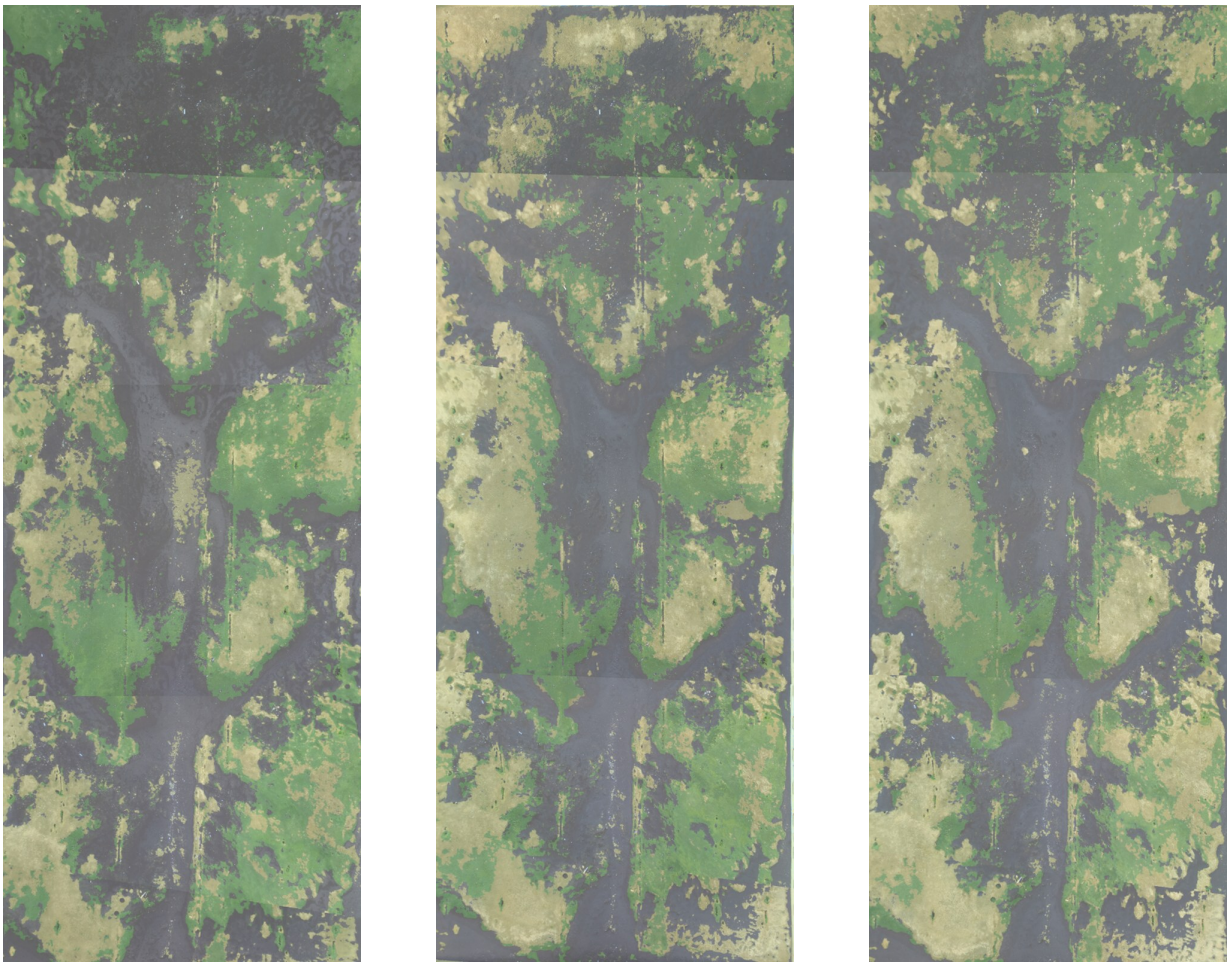
The goal of this note is to determine how stable moss occurrence is related to bath geometry and derived topographic variables. The central question is not only where water accumulates, but why the strongest moss signal is not observed in the deepest part of the central trough.

The analysis uses three stitched observations of the same LEO1 bath, acquired on 3 November 2022, 22 November 2022, and 7 December 2022. These images were interpreted together with the soil-surface raster `LEO1_soil.asc` and the container-geometry raster `LEO1_container.asc`. The image data were rectified to the bath coordinate system before comparison with the topographic rasters.

The working hypothesis tested here is that moss responds to a moisture window, not to wetness monotonically. In this interpretation, the central trough may be wet, but it may also be too saturated, too frequently flushed, or too unstable to support persistent moss growth.

## 2 Image Registration and Moss Proxy

The three RGB mosaics were cropped to the soil surface, corrected by a perspective transform, and rescaled to a common rectangular domain. This step removes the surrounding frame and makes the three dates directly comparable in bath coordinates.



(a) 3 November 2022 (b) 22 November 2022 (c) 7 December 2022

Figure 1: Rectified source images used in the analysis. The large-scale moss distribution is already visible on the first date and remains spatially coherent through December. The later images show stronger green coloration, consistent with seasonal greening of an existing pattern.

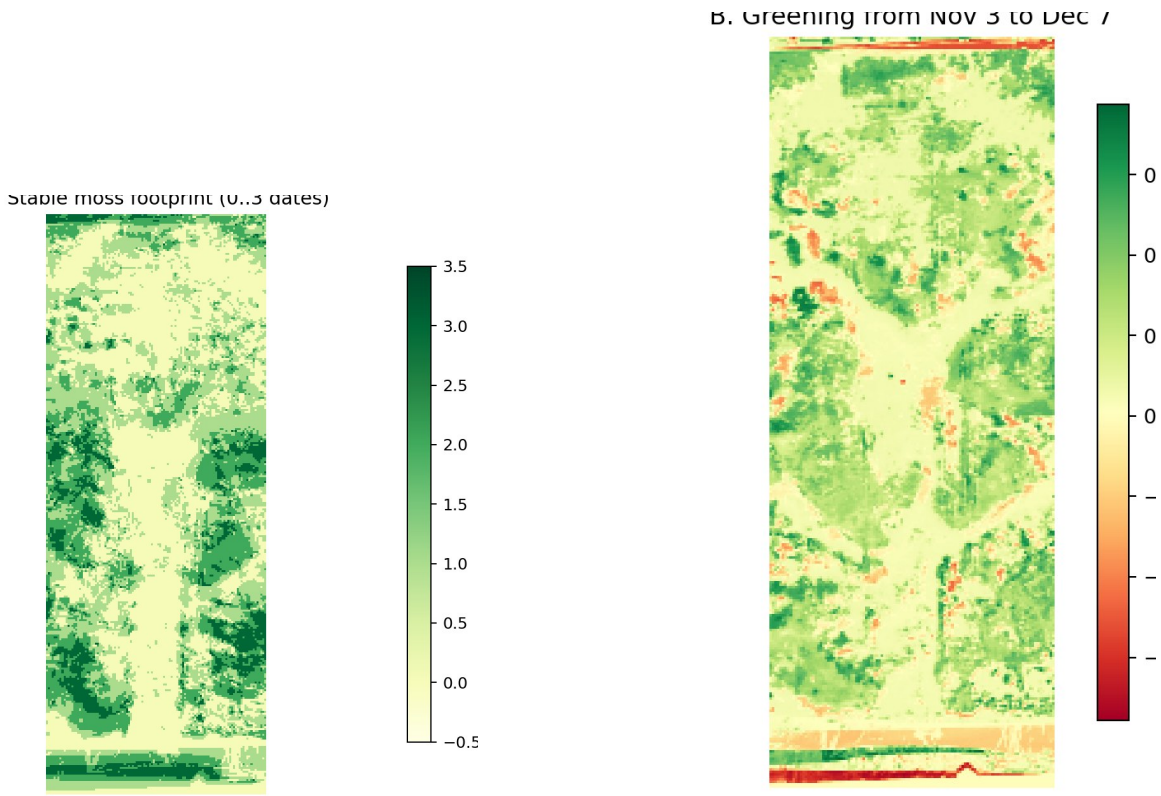
A broad moss proxy was used instead of a narrow green threshold. This choice is important because moss color changes from yellow-green to greener tones over the observation period. The proxy therefore aims to represent moss presence rather than a single color state. From the date-specific proxies, two derived quantities were computed:

1. a persistent moss footprint, defined by repeated occurrence across the three dates;
2. a seasonal greening signal, defined by the change between early November and early December.

The persistent footprint is the primary response variable in the analysis below. It is compared with lateral position $|x|$, downstream coordinate $s$, trough depth, a flow-accumulation proxy, wall proximity, and soil thickness relative to the container.

## 3 Observed Moss Pattern

The persistent moss footprint is broad and structured. It is not concentrated in the deepest central corridor. Instead, the main stable patches occur on the flanks and in the outer half of the bath.




(a) Persistent moss footprint

(b) Seasonal greening

Figure 2: Moss-centered diagnostics. The stable moss footprint is displaced from the deepest central trough, whereas the strongest seasonal greening is largely co-located with regions that were already moss-bearing. This supports the interpretation of seasonal color change rather than rapid colonization of entirely new zones.

Table 1 summarizes the main spatial statistics. The median lateral position of stable moss is 3.7 m, placing the stable moss core well away from the centerline. The corresponding median trough depth is only 0.088 m, indicating that the stable zone is associated with shallow to intermediate topographic depression rather than the deepest drainage core.

Table 1: Summary metrics for the stable moss pattern.

| Metric | Value |
|---|---|
| Cells with stable moss signal, present on at least two dates | 26.6% |
| Cells strong on all three dates | 7.1% |
| Median lateral position of the stable moss zone, $\lvert x \rvert$ | 3.7 m |
| Median trough depth of the stable moss zone | 0.088 m |
| Median $\log(1 + \text{flow})$ of the stable moss zone | 2.20 |
| Median downstream position of the stable moss zone | 17.65 m |

## 4 Dendritic Moss-Free Pattern and Topographic Controls

The rectified photographs of the LEO1 surface show that regions of weak or absent moss are not randomly distributed. Instead, they form a connected dendritic pattern that resembles a surface drainage network. This pattern is consistent with preferential flow over an inclined heterogeneous substrate: small topographic depressions collect runoff, and neighboring depressions become connected into larger rills and tributary-like channels.

The central dark corridor and its lateral branches therefore appear to be negative biological imprints of the most hydrologically active parts of the surface. Although these corridors are likely wetter than the surrounding shoulder zones, they need not provide the most favorable habitat for moss. Concentrated flow can suppress persistent moss cover through several coupled mechanisms:

1. **Excess saturation.** Persistent water saturation can reduce oxygen availability in the near-surface layer and create unfavorable conditions for moss growth.
2. **Surface flushing.** Concentrated runoff can remove spores, fine particles, organic material, and weakly attached biological crust.
3. **Sediment redistribution.** Flowing water can repeatedly transport and redeposit fine sediment along active drainage paths, thereby disturbing the colonized surface.
4. **Mechanical instability.** Active flow corridors may undergo more frequent surface reorganization than adjacent shoulders, reducing the likelihood of stable moss establishment.
5. **Moisture-window limitation.** Moss growth is favored not by maximal wetness, but by an intermediate moisture regime. The flanks of the drainage network can remain sufficiently moist while avoiding prolonged saturation and strong flushing.

Thus the dendritic moss-free pattern is interpreted as a drainage-controlled exclusion zone rather than as a primary biological branching structure. In this interpretation, moss-rich regions occupy the shoulders and flanks of the drainage network, whereas moss-poor regions mark the most active flow corridors. The photographs therefore support a bounded-moisture hypothesis: persistent moss cover is favored where water availability is sufficient but not extreme, while the deepest and most actively draining portions of the central trough are too wet, too unstable, or too frequently flushed to support sustained colonization.

## 5 Topographic Response Structure

The response curves in Fig. 3 provide quantitative support for this interpretation. The persistent moss proxy increases with lateral distance from the centerline, indicating that moss cover is displaced away from the central trough. Its dependence on trough depth is non-monotone: moss is favored at shallow to intermediate depths and decreases sharply once the trough becomes sufficiently deep. In addition, large values of the flow-accumulation proxy are associated with weaker moss signal.

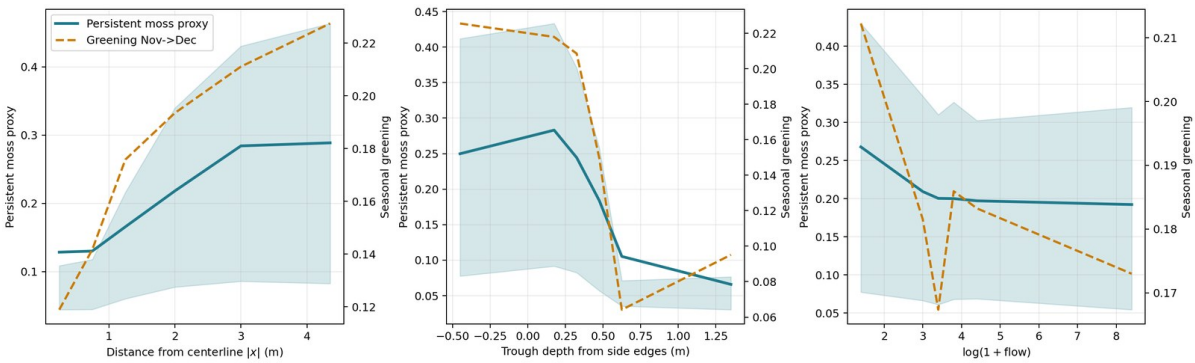



Figure 3: Response curves for persistent moss and seasonal greening. The persistent moss proxy increases with distance from the centerline, reaches its largest values at shallow to intermediate trough depths, and decreases in the deepest trough. The decline beyond approximately 0.4 m provides evidence against a monotone "wetter is better" interpretation.

The same spatial niche is visible in bath coordinates (Fig. 4). The highest mean persistent moss signal occurs in the outer half of the bath, especially for $|x| \gtrsim 3\,\mathrm{m}$, and is most pronounced in the middle and lower portions of the domain.

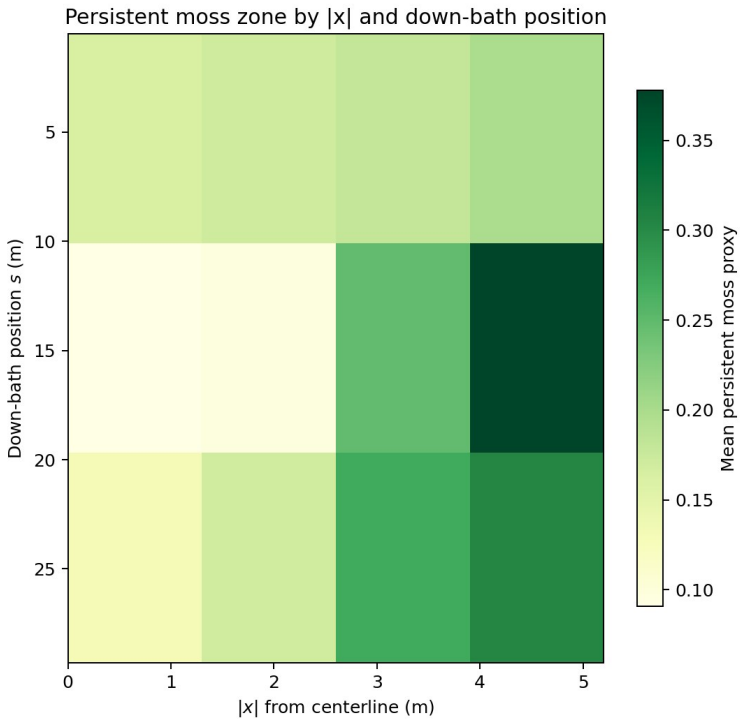



Figure 4: Mean persistent moss proxy as a function of lateral position $|x|$ and downstream coordinate $s$. The highest values occur away from the centerline and are concentrated in the middle and lower portions of the bath.

# 6 Cross-Sectional Evidence

Cross-sectional profiles provide the clearest geometric test of the proposed interpretation. In several sections, the elevation minimum of the central trough is moss-poor, whereas higher moss values occur on the flanks and outer shoulders. This demonstrates that the observed relationship is not merely an artifact of two-dimensional map overlay or color thresholding.

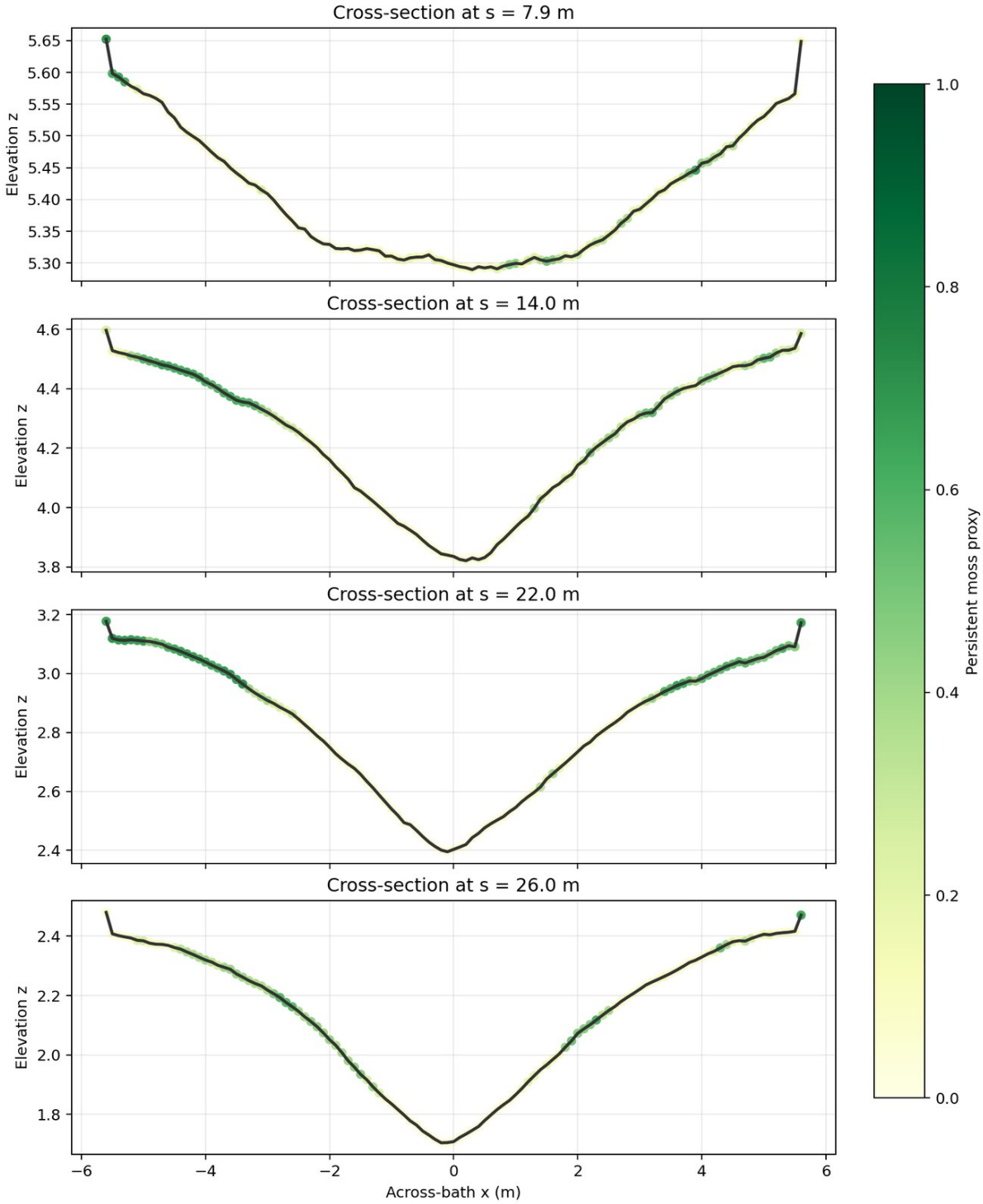



Figure 5: Cross-sections through the bath surface colored by the persistent moss proxy. The trough minima are repeatedly moss-poor, whereas stronger moss signal occurs on the flanks and outer shoulders. This directly supports the shoulder-zone interpretation.

Together, the photographic, response-curve, and cross-sectional evidence indicate that the stable moss habitat is not located in the deepest part of the central drainage corridor. Instead, it is displaced toward off-channel shoulder zones where moisture is available but hydrological disturbance is weaker.

## 7 Smooth Surface Representation

For geometric analysis, the soil surface was represented by a cubic bivariate smoothing spline,

$$z(x, s) \approx S(x, s),$$

fitted to the valid cells of `LEO1_soil.asc`. The reported root-mean-square error of the smoothed surface is approximately

$$\mathrm{RMSE} \approx 0.071\,\mathrm{m}.$$

The spline is used as a differentiable representation of the raster topography, not as a physical seepage model. Its purpose is to make quantities such as local slope, trough depth, and cross-sectional profile easier to interpret.

## 8 Stable Moss Ranking Model

Stable moss occurrence was summarized with a compact logistic ranking model,

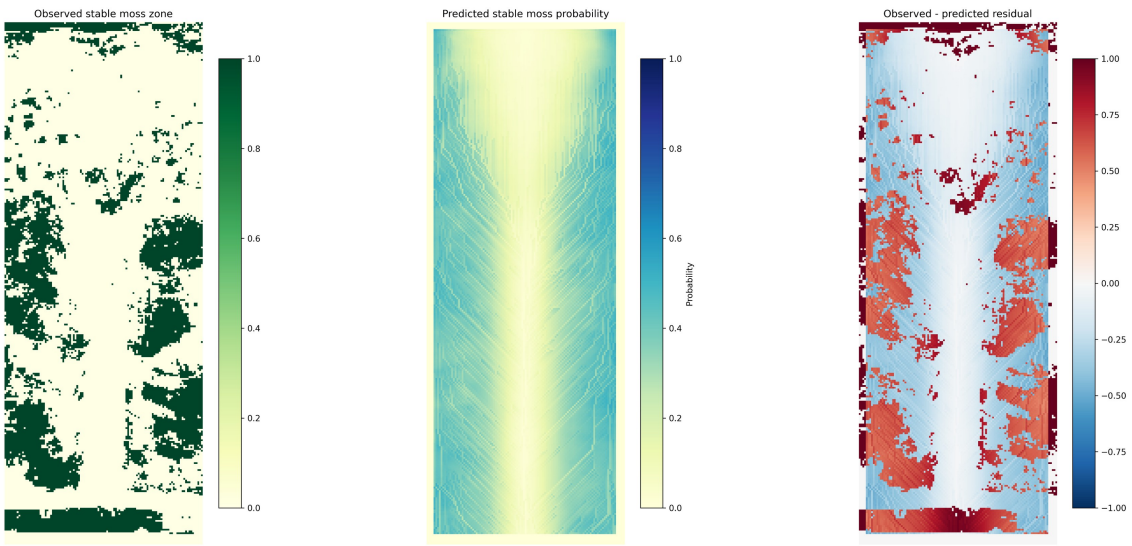
$$\operatorname{logit} P(M = 1) = \beta_0 + \beta_w W - \beta_f F - \gamma(d - d_0)^2 + \beta_t T, \tag{1}$$

where $W$ is wall proximity, $F$ is standardized $\log(1 + \text{flow})$, $d$ is trough depth, and $T$ is standardized soil thickness relative to the container. The quadratic depth term represents a bounded moisture window: the probability of stable moss is allowed to peak at an intermediate depth and to decline when the trough becomes too shallow or too deep.

Table 2: Fitted parameters of the stable moss ranking model.

| Parameter | Estimate |
|---|---:|
| $\beta_0$ | $-0.307$ |
| $\beta_w$ | 1.286 |
| $\beta_f$ | 0.185 |
| $\gamma$ | 11.971 |
| $d_0$ | 0.407 m |
| $\beta_t$ | 0.146 |

The model is not intended as a complete mechanistic law. It is a low-dimensional ranking model that tests whether topographic variables can order favorable and unfavorable regions. Its performance is moderate but meaningful: the reported training AUC is 0.724 and the spatial cross-validated AUC is 0.657.




(a) Observed stable moss (b) Predicted probability (c) Observed minus predicted

Figure 6: Spatial diagnostics for the stable moss ranking model. The model reproduces the broad displacement of stable moss away from the central hollow, although residual structure indicates that additional biological or hydrological variables remain unresolved.

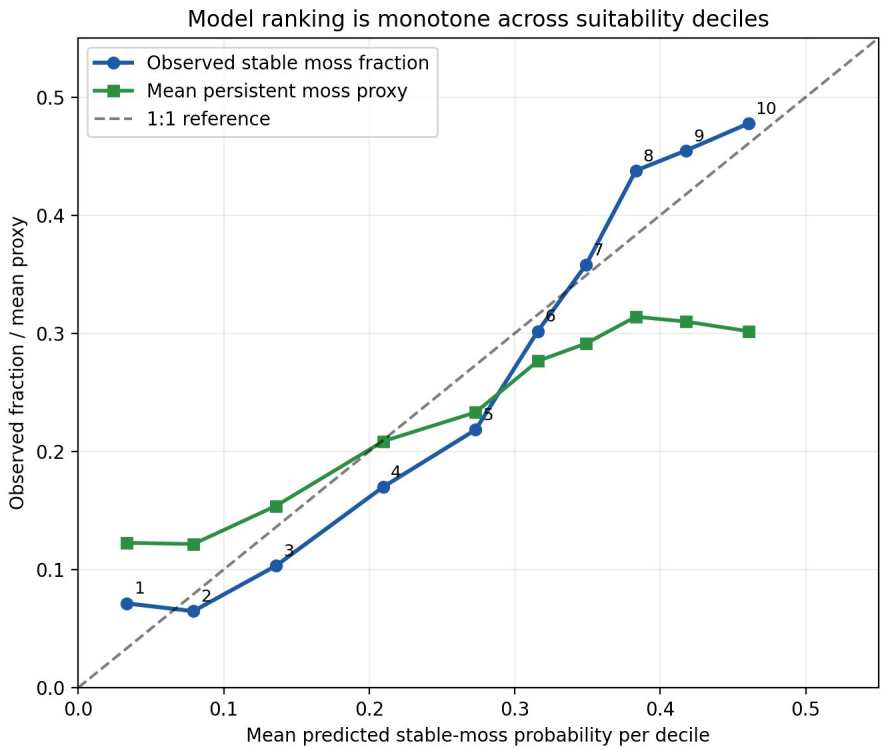
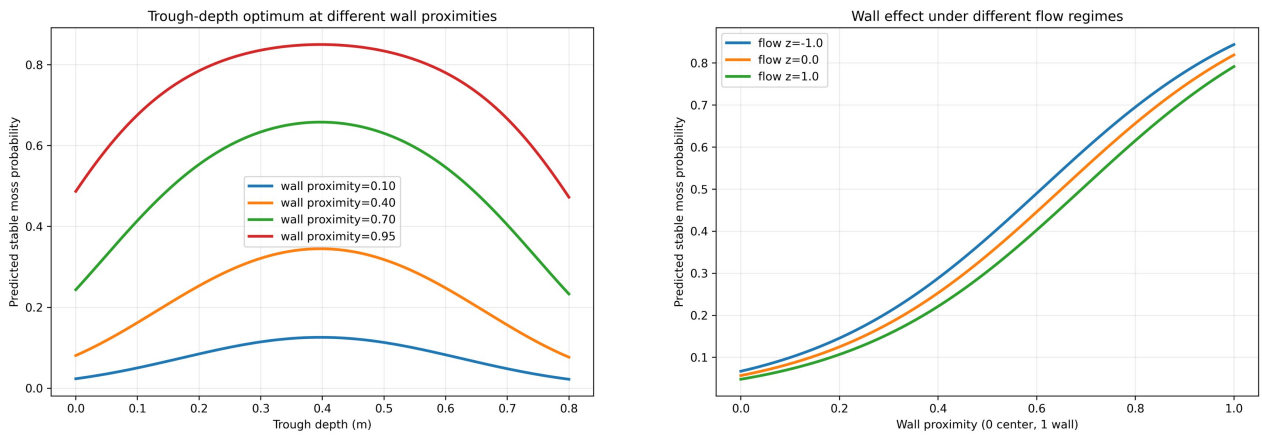



Figure 7: Calibration of the ranking model. Across predicted-suitability deciles, the observed stable-moss fraction increases from about 7% to about 48%. The monotone ordering supports the use of the model as a suitability ranking, even though it is not a full process model.




(a) Trough-depth response

(b) Wall-proximity response

Figure 8: One-dimensional model responses. The depth curves exhibit a non-monotone optimum near 0.4 m, while wall proximity increases the predicted probability over the range shown. Increasing flow shifts suitability downward.

## 9 Biophysical Interpretation

The observed pattern is most naturally interpreted as evidence for a bounded moisture optimum. The central trough is likely wetter than the flanks, but it may also experience longer saturation, lower oxygen availability, sediment redistribution, flushing, and more frequent surface disturbance. These effects can make the wettest zone less favorable biologically, even if water is not limiting there.

The flanks of the trough provide a plausible intermediate regime. They remain moist but are less likely to experience persistent waterlogging or strong surface washout. The positive wall-proximity term in the ranking model suggests that the box-like container geometry may further stabilize these outer shoulder regions, either by retaining moisture laterally or by reducing erosive surface flow.

This interpretation is consistent with broader observations from moss and biological soil-crust systems: moisture promotes activity only up to an optimum, whereas excessive saturation, hypoxic stress, runoff, and disturbance can suppress growth or persistence [1, 2, 3, 4, 5].

## 10 Limitations and Needed Measurements

The present analysis establishes a spatial association between moss persistence and bath geometry. It does not directly measure the water state, oxygen status, or disturbance history of the substrate. A mechanistic test of the bounded moisture hypothesis would require at least one of the following:

- local time series of soil moisture or matric potential;
- observations of standing water or saturation duration in the trough;
- tracer experiments to quantify flushing and residence time;
- repeated high-resolution surveys after controlled irrigation events;
- measurements of fine-sediment redistribution and surface stability.

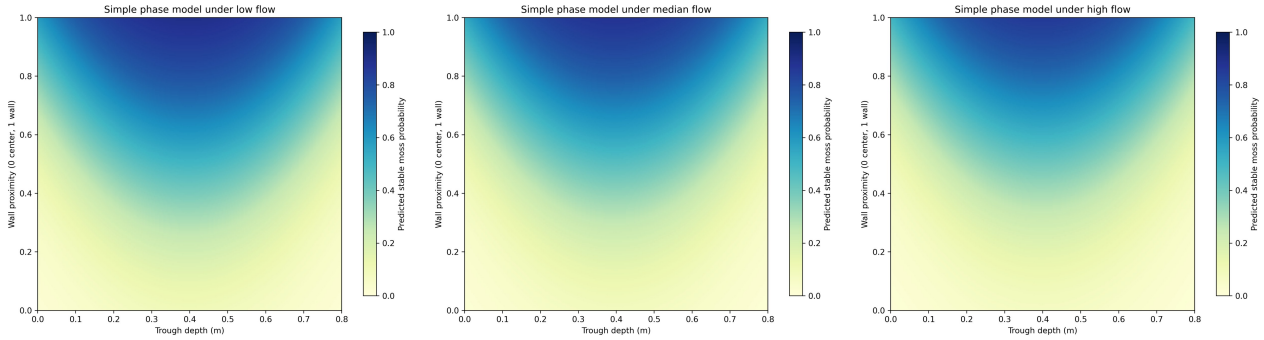
These measurements would distinguish between two related mechanisms: excessive wetness itself and hydrologic instability caused by concentrated flow.

## 11 Conclusion

The dominant biological pattern in the LEO1 bath is not simple occupation of the wettest available ground. The stable moss footprint is displaced away from the deepest central trough and is concentrated on off-center shoulder zones. The combined evidence from rectified imagery, response curves, cross-sections, and a compact ranking model supports a bounded-moisture interpretation: moss is favored where the substrate remains sufficiently moist but is not maximally wet, strongly flushed, or hydrologically unstable.

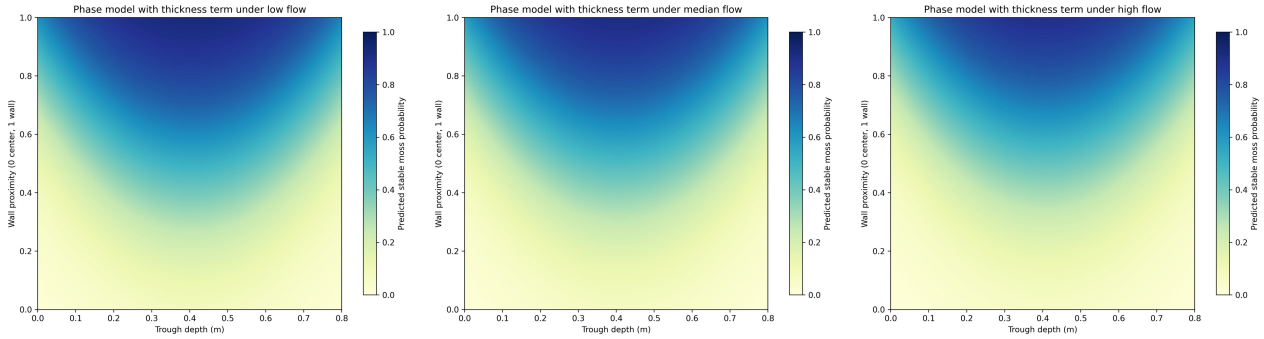
## A Additional Phase-Model Surfaces

The following panels show the model probability surfaces under different flow conditions. They are included as supporting diagnostics; the main text relies primarily on the observed footprint, cross-sections, response curves, and one-dimensional model responses.




(a) Low flow (b) Median flow (c) High flow

Figure 9: Simple phase-model suitability surfaces under increasing flow.




(a) Low flow (b) Median flow (c) High flow

Figure 10: Phase-model suitability surfaces with the soil-thickness term. The additional term modestly changes the ranking but does not alter the main interpretation: stable moss is favored away from the deepest central trough.

## References

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